

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396725
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Wahl; Rebecca Hawkins

Surgical bending instrument

Abstract

An apparatus and a method are provided for a surgical bending instrument for bending surgical implants. The surgical bending instrument comprises a body including a longitudinally extending threaded hole. A shaft comprising a threaded portion is rotatably engaged within the threaded hole. A handle is coupled to a proximal end of the shaft, and a distal extension of the shaft comprises a driver. A distal force applicator comprises the driver centered between a first grip and a second grip. In some embodiments, the distal force applicator is configured to retain a surgical staple, such that the surgical staple may be changed to a distracted configuration suitable for implantation at a bone fixation or fusion site of a patient. In some embodiments, the distal force applicator is configured to bend a bone fusion plate so as to tailor the plate to specific anatomy of the patient's bone.

Inventors:	Wahl; Rebecca Hawkins (Escondido, CA)
Applicant:	In2Bones USA, LLC (Memphis, TN)
Family ID:	1000008780294
Assignee:	In2Bones USA, LLC (Memphis, TN)
Appl. No.:	18/117804
Filed:	March 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230200809 A1	Jun. 29, 2023

Related U.S. Application Data

continuation parent-doc US 15992065 20180529 US 10863982 child-doc US 18117804
continuation parent-doc US 17100711 20201120 US 11596398 child-doc US 18117804

Publication Classification

Int. Cl.: **A61B17/068** (20060101); **A61B17/064** (20060101); **A61B17/072** (20060101);
A61B17/88 (20060101)

U.S. Cl.:

CPC **A61B17/0682** (20130101); **A61B17/064** (20130101); **A61B17/0642** (20130101);
A61B17/072 (20130101); **A61B17/8863** (20130101); A61B2017/0641 (20130101);
A61B2017/0645 (20130101); A61B2017/07235 (20130101)

Field of Classification Search

CPC: A61B (17/0682)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2009/0192538	12/2008	Sandel	606/167	A61B 17/3213
2016/0199060	12/2015	Morgan	227/175.1	A61B 17/068

Primary Examiner: Sevilla; Christian A

Attorney, Agent or Firm: Rutan & Tucker LLP

Background/Summary

PRIORITY (1) This application is a continuation, and claims the benefit of, U.S. patent application Ser. No. 17/100,711 filed Nov. 20, 2020, which claims the benefit of, U.S. patent application Ser. No. 15/992,065 filed May 29, 2018 and U.S. Patent Application, entitled “Surgical Bending Instrument,” filed on Apr. 6, 2017 and having application Ser. No. 15/480,612, which is a continuation of, and claims the benefit of, U.S. Patent Application, entitled “Surgical Bending Instrument,” filed on Feb. 11, 2016 and having application Ser. No. 15/041,959, which claims the benefit of, and priority to, U.S. Provisional Application, entitled “Surgical Bending Instrument,” filed on Feb. 14, 2015 and having application Ser. No. 62/116,387, the entirety of each of said applications being incorporated herein by reference.

FIELD

(1) The field of the present disclosure generally relates to securing bones together. More particularly, the field of the disclosure relates to an apparatus and a method for a surgical bending instrument for implanting surgical implants at bone fixation or fusion sites of a patient.

BACKGROUND

(2) Surgical staples or a fusion bone plate implant utilized in conjunction with one or more fasteners may be used to generate compression and stability at a bone interface. An implant

generally serves to stabilize bones, or bone parts, relative to one another so as to promote bone fusion. In many applications, surgical staples, or bone plates and fasteners are used to fuse bones, or bone parts, of the human body, such as bones in the foot, the ankle, the hand, the wrist, as well as various other portions of the body. Surgical staples are particularly advantageous in the hands and feet due to a low dorsal profile of the staples once they are implanted. Furthermore, during the course of certain medical procedures, a surgeon may immobilize one or more bones or bone fragments by stabilizing the bones together in a configuration which approximates the natural anatomy. To this end, the surgeon may use fasteners to attach the bones to a bone plate implant so as to hold the bones in alignment with one another while they fuse together.

SUMMARY

(3) An apparatus and a method are provided for a surgical bending instrument for bending surgical implants. The surgical bending instrument comprises a body including a longitudinally extending threaded hole. A shaft comprising a threaded portion is rotatably engaged within the threaded hole. A handle is coupled to a proximal end of the shaft, and a distal extension of the shaft comprises a driver. A distal force applicator comprises the driver centered between a first grip and a second grip. In some embodiments, the distal force applicator is configured to retain a surgical staple, such that the surgical staple may be changed to a distracted configuration suitable for implantation at a bone fixation or fusion site of a patient. In some embodiments, the distal force applicator is configured to bend a bone fusion plate so as to tailor the plate to specific anatomy of the patient's bone. The surgical bending instrument may be packaged into sterile surgery-specific kits comprising other surgical tools, such as drill guides, drill sizers, tamps, forceps, staple removal tools, drills, temporary pins, drill depth stops, fusion bone plates, bone plate fasteners, compression screws, and the like.

(4) In an exemplary embodiment, a surgical bending instrument for bending surgical implants comprises a body supporting a distal force applicator and a proximal handle; a first grip and a second grip of the distal forced applicator configured to retain the surgical implant; and a driver of the distal force applicator configured to exert a distally-directed force so as to bend the surgical implant.

(5) In another exemplary embodiment, a shaft is threadably engaged within a longitudinally extending hole within the body, a distal extension of the shaft comprising the driver, and a proximal handle is fixedly coupled to a proximal end of the shaft. In another exemplary embodiment, the driver and the shaft are separate components that are engaged with one another. In another exemplary embodiment, the shaft communicates mechanical forces applied at the proximal handle to the driver. In another exemplary embodiment, rotating the proximal handle relative to the body moves the driver longitudinally relative to the body. In another exemplary embodiment, twisting the proximal handle clockwise moves the driver distally into contact with the crown, such that the surgical staple is clasped between the grips and the driver, wherein further clockwise twisting of the proximal handle changes the surgical staple from an initial configuration to a distracted configuration suitable for implantation across a bone fusion or fixation site of a patient. In another exemplary embodiment, twisting the proximal handle counterclockwise retracts the driver proximally away from the crown, allowing the surgical staple to relax from the distracted configuration and compress the bone fusion or fixation site.

(6) In another exemplary embodiment, a lever comprising a wheel and a peripheral cam is rotatably mounted on a pivot within the body, such that when the lever is moved proximally, the cam forcibly pushes the driver so as to exert a distally-directed force onto the surgical implant. In another exemplary embodiment, a lever comprising a wheel and a peripheral cam is rotatably mounted on a pivot within the body, such that moving the lever proximally causes the cam to exert a distally-directed force onto the surgical implant. In another exemplary embodiment, the surgical bending instrument comprises a rigid material suitable for bending surgical implants.

(7) In another exemplary embodiment, the distal force applicator is configured to retain a surgical

staple, such that the surgical staple may be changed to a distracted configuration suitable for implantation at a bone fixation or fusion site of a patient. In another exemplary embodiment, the surgical staple is indicated for fixation of osteotomies and joint arthrodeses of the hands and feet. In another exemplary embodiment, the distal force applicator is configured to bend a bone fusion plate so as to tailor the plate to specific anatomy of a patient's bone.

(8) In another exemplary embodiment, the first and second grips are configured to support a crown of a surgical staple when the driver is placed into forcible contact with the center of the crown. In another exemplary embodiment, the driver and the grips operate to retain the surgical staple within the distal force applicator, thereby facilitating implanting the staple into the patient. In another exemplary embodiment, the first and second grips engage the surgical staple on opposite sides of the crown, thereby preventing the surgical staple from becoming dislodged from the distal force applicator during implantation into a patient. In another exemplary embodiment, the first and second grips have a separation distance comparable with the length of the crown. In another exemplary embodiment, the first and second grips have an adjustable separation distance so as to facilitate using the surgical bending instrument with a variety of differently-sized staples.

(9) In another exemplary embodiment, the surgical bending instrument is packaged into sterile surgery-specific kits comprising other surgical tools, such as drill guides, drill sizers, tamps, forceps, staple removal tools, drills, temporary pins, drill depth stops, fusion bone plates, bone plate fasteners, compression screws, and the like.

(10) In an exemplary embodiment, a method of using a surgical bending instrument for implanting a surgical staple at a bone fixation or fusion site of a patient comprises loading the surgical staple into a distal force applicator of the surgical bending instrument, such that the surgical staple is in contact with a first grip and a second grip; advancing a driver into forcible contact with a crown of the surgical staple, the surgical staple being clasped between the first grip, the second grip, and the driver; distracting the surgical staple, such that a first leg and a second leg of the surgical staple are parallel to one another; inserting the surgical staple through an incision and sliding the first and second legs into parallel holes drilled across a bone fusion or fixation site of the patient; retracting the driver from the crown so as to allow the surgical staple to compress the bone fusion or fixation site; disengaging the first and second grips from the crown; pushing the surgical staple into the parallel holes until the crown is in contact with the patient's bone; and closing the incision by way of suturing.

(11) In another exemplary embodiment, advancing further comprises rotating a proximal handle clockwise so as to move the driver distally into contact with the crown, the driver being connected to the proximal handle by way of a threaded shaft extending longitudinally through the surgical bending instrument. In another exemplary embodiment, distracting the surgical staple further comprises twisting the proximal handle clockwise to move the driver distally into the crown until the first and second legs are parallel with one another. In another exemplary embodiment, retracting further comprises rotating the proximal handle counterclockwise so as to move the driver proximally away from the crown, thereby allowing the first and second legs to bend toward one another.

(12) In an exemplary embodiment, a surgical bending instrument for implanting a surgical staple at a bone fusion site of a patient comprises: a body supporting a first grip and a second grip that are configured to retain a crown of the surgical staple; a driver configured to exert a distally-directed force onto the top of the crown between the first grip and the second grip; and a proximal handle configured to move the driver longitudinally relative to the body.

(13) In another exemplary embodiment, the first grip and the second grip are disposed on the same side of the surgical staple and configured to support opposite ends underneath the crown. In another exemplary embodiment, the first grip and the second grip are configured to forcibly retain the crown when the driver contacts the center of the crown. In another exemplary embodiment, the first grip and the second grip have a separation distance comparable with the length of the crown. In

another exemplary embodiment, the first grip and the second grip have an adjustable separation distance so as to facilitate retaining a variety of differently-sized surgical staples.

(14) In another exemplary embodiment, a shaft extending from the proximal handle to the driver is threadably engaged within the body, such that rotating the proximal handle moves the driver relative to the body. In another exemplary embodiment, clockwise rotation of the proximal handle moves the driver distally and clasps the crown between the first grip, the second grip, and the driver, such that further clockwise rotation places the surgical staple into a distracted configuration suitable for implantation across the bone fusion site. In another exemplary embodiment, counterclockwise rotation of the proximal handle moves the driver proximally, allowing the surgical staple to relax from the distracted configuration and compress the bone fusion site.

(15) In another exemplary embodiment, the proximal handle comprises a lever coupled with a cam that is pivotally mounted within the body, such that proximal movement of the lever causes the cam to push the driver into forcible contact with the crown. In another exemplary embodiment, moving the lever to a maximally proximal position places the surgical staple into a distracted configuration suitable for implantation across the bone fusion site. In another exemplary embodiment, moving the lever distally allows the surgical staple to relax from the distracted configuration and compress the bone fusion site.

(16) In an exemplary embodiment, a surgical bending instrument for distracting a surgical staple for implantation at a bone fusion site of a patient comprises: a first grip and a second grip that are disposed on the same side of the surgical staple and configured to support opposite ends underneath a crown of the surgical staple; a driver in mechanical communication with a proximal handle and configured to exert a distal force on top of the crown; and a body supporting at least the first grip, the second grip, and the driver.

(17) In another exemplary embodiment, the proximal handle is configured to facilitate moving the driver distally to clasp the crown between the first grip, the second grip, and the driver, and thereby place the surgical staple into a distracted configuration suitable for implantation across the bone fusion site, and wherein the proximal handle is configured to move the driver proximally to allow the surgical staple to relax from the distracted configuration and compress the bone fusion site after implantation. In another exemplary embodiment, the proximal handle is coupled with the driver by way of a shaft threadably retained within the body, whereby rotating the proximal handle moves the driver longitudinally with respect to the body. In another exemplary embodiment, the proximal handle is coupled with the driver by way of a cam that is mounted pivotally within the body, whereby moving a lever coupled with the cam moves the driver longitudinally relative to the body.

(18) In an exemplary embodiment, a surgical staple for fixating osteotomies and joint arthrodeses comprises a crown comprising an elongate member having a first end and a second end; a first leg coupled with the first end and extending obliquely from the crown toward a centerline of the crown; a second leg coupled with the second end and extending obliquely from the crown toward the centerline of the crown, the first leg and the second leg extending toward one another and sharing an acute angle; and a plurality of teeth disposed on an inner side of the first leg and an inner side the second leg, the plurality of teeth being configured to engage with a bone.

(19) In another exemplary embodiment, the surgical staple is comprised of a material that is amenable to being placed into a distracted configuration and relaxed therefrom, such as by way of a suitable surgical bending instrument. In another exemplary embodiment, the material is comprised of a metal alloy that exhibits shape memory and superelastic properties, such as Nitinol or other similar material.

(20) In another exemplary embodiment, the plurality of teeth are biased toward the crown, such that the plurality of teeth offer relatively little resistance during implantation of the surgical staple into the bone and fixate the surgical staple within the bone after implantation. In another exemplary embodiment, the plurality of teeth is comprised of seven teeth disposed along each of the first leg and the second leg. In another exemplary embodiment, one or more exterior teeth are disposed

along each of the first leg and the second leg, the one or more exterior teeth being configured to cooperate with the plurality of teeth to engage the bone and prevent movement of the surgical staple after being implanted into the bone. In another exemplary embodiment, one or more teeth are disposed along the crown and configured to engage with bone and prevent lateral movement of the crown along the surface of the bone. In another exemplary embodiment, the one or more teeth are biased toward a centerline of the crown.

(21) In another exemplary embodiment, the crown further comprises an upper reinforced portion and a valley underneath the reinforced portion, the valley being centrally disposed along the lower surface of the crown and bordered by edges. In another exemplary embodiment, the edges are configured to engage with bone and prevent lateral movement of the crown along the surface of the bone once the surgical staple is implanted. In another exemplary embodiment, the reinforced portion is configured to impart structural integrity to the surgical staple in compensation for the presence of the valley.

(22) In another exemplary embodiment, the crown is comprised of wing portions disposed at opposite ends of the crown and extending beyond the first leg and the second leg. In another exemplary embodiment, the wing portions are configured to cooperate with at least the plurality of teeth to fixate the surgical staple once implanted in the bone. In another exemplary embodiment, the wing portions are configured to receive grips of a surgical bending instrument that is configured to place the surgical staple into a distracted configuration.

(23) In an exemplary embodiment, a surgical staple for fixating osteotomies and joint arthrodeses comprises a crown comprised of an elongate member having a first end and a second end; a first transverse portion disposed at the first end; a second transverse portion disposed at the second end; a first leg and a second leg extending downward and parallelly from opposite ends of the first transverse portion; a third leg and a fourth leg extending downward and parallelly from opposing ends of the second transverse portion; and a plurality of teeth disposed on each of the first leg, the second leg, the third leg, and the fourth leg and oriented toward a centerline of the crown.

(24) In another exemplary embodiment, the first leg and the third leg extend obliquely from the crown toward the centerline and share an intervening acute angle, and wherein the second leg and the fourth leg extend obliquely from the crown toward the centerline and share an intervening acute angle. In another exemplary embodiment, the plurality of teeth disposed on the first leg and the plurality of teeth disposed on the third leg are oriented toward one another and biased toward the crown, such that the first leg and the third leg cooperate to engage with a bone disposed therebetween. In another exemplary embodiment, the plurality of teeth disposed on the second leg and the plurality of teeth disposed on the fourth leg are oriented toward one another and biased toward the crown, such that the second leg and the fourth leg cooperate to engage with a bone disposed therebetween.

(25) In another exemplary embodiment, one or more exterior teeth are disposed along at least one of the first leg, the second leg, the third leg, and the fourth leg, the one or more exterior teeth being configured to cooperate with the plurality of teeth to engage a bone and prevent movement of the surgical staple after being implanted into the bone. In another exemplary embodiment, wing portions are disposed at opposite ends of the crown and extending beyond the first transverse portion and the second transverse portion. In another exemplary embodiment, the wing portions are configured to receive grips of a surgical bending instrument that is configured to place the surgical staple into a distracted configuration.

(26) In an exemplary embodiment, a surgical bending instrument for implanting a surgical staple at a bone fusion site of a patient comprises: a body having a distal force applicator and a proximal handle; a first grip and a second grip comprising the distal force applicator; and a driver in mechanical communication with the proximal handle.

(27) In another exemplary embodiment, the first grip and the second grip are separated by a distance comparable with a length of a crown comprising the surgical staple. In another exemplary

embodiment, the first grip is configured to be positioned near a first leg of the surgical staple while the second grip is configured to be positioned near a second leg of the surgical staple. In another exemplary embodiment, the distal force applicator is configured to provide an adjustable separation distance between the first grip and the second grip.

(28) In another exemplary embodiment, the distal force applicator is configured to be used with specific sizes of surgical staples. In another exemplary embodiment, any one or more of the body, the distal force applicator, and the proximal handle are configured with an overall size so as to be used with specific sizes of the surgical staple. In another exemplary embodiment, any one or more of the body, the distal force applicator, and the proximal handle include a color-matching system so as to indicate those certain surgical staples that are implantable by way of the surgical bending instrument.

(29) In another exemplary embodiment, a shaft is disposed inside the body and extends from the proximal handle to the distal force applicator. In another exemplary embodiment, the driver comprises a distal extension of the shaft that is configured to forcibly contact a crown comprising the surgical staple. In another exemplary embodiment, the shaft includes a threaded portion configured to advance the driver into the crown when the proximal handle is turned with respect to the body.

(30) In an exemplary embodiment, a surgical bending instrument for implanting a surgical staple at a bone fusion site of a patient comprises: a body having a distal force applicator and a lever; a first grip and a second grip comprising the distal force applicator; and a driver in mechanical communication with the lever.

(31) In another exemplary embodiment, the lever comprises a wheel that is rotatably mounted on a pivot within the body and is in sliding contact with a proximal portion of the driver. In another exemplary embodiment, the wheel includes a cam configured to push the driver distally relative to the body when the lever is moved from a first position to a second position. In another exemplary embodiment, any one or more of the lever, the wheel, and the cam are configured to cause the driver to distract the surgical staple when the lever is in the second position. In another exemplary embodiment, the body includes a narrow midsection and a proximal surface that are configured to facilitate grasping and stabilizing the surgical bending instrument during moving the lever from the first position to the second position.

(32) In another exemplary embodiment, the driver comprises a cam that is disposed on a wheel comprising the lever, the wheel being rotatably mounted on a pivot within the body. In another exemplary embodiment, the cam is centrally disposed within the body so as to contact a crown of the surgical staple between the first grip and the second grip. In another exemplary embodiment, any one or more of the lever, the wheel, and the cam are configured to distract the surgical staple when the lever moved from a first position to a second position. In another exemplary embodiment, the lever includes a slot that is configured to accept the crown when the lever is moved into the first position. In another exemplary embodiment, the slot is configured to cooperate with the first grip and the second grip to retain the surgical staple in the surgical bending instrument while the lever is in the first position.

(33) These and other features of the concepts provided herein may be better understood with reference to the drawings, description, and appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The drawings refer to embodiments of the present disclosure in which:
- (2) FIG. 1 illustrates an upper perspective view of an exemplary embodiment of a surgical bending instrument in accordance with the present disclosure;

- (3) FIG. 2 illustrates a lower perspective view of an exemplary embodiment of a distal force applicator, according to the present disclosure;
- (4) FIG. 3A illustrates a ghost view of an exemplary embodiment of a surgical bending instrument retaining a surgical staple in an initial configuration, according to the present disclosure;
- (5) FIG. 3B illustrates a ghost view of the exemplary embodiment of the surgical bending instrument illustrated in FIG. 3A distracting the surgical staple in accordance with the present disclosure;
- (6) FIG. 4A illustrates a ghost view of an exemplary embodiment of a surgical bending instrument retaining a surgical staple in an initial configuration, according to the present disclosure;
- (7) FIG. 4B illustrates a ghost view of the exemplary embodiment of the surgical bending instrument illustrated in FIG. 4A distracting the surgical staple in accordance with the present disclosure;
- (8) FIG. 5A illustrates an upper perspective view of an exemplary embodiment of a surgical bending instrument retaining a surgical staple in an initial configuration in accordance with the present disclosure;
- (9) FIG. 5B illustrates an upper perspective view of the exemplary embodiment of the surgical bending instrument illustrated in FIG. 5A distracting the surgical staple in accordance with the present disclosure;
- (10) FIG. 6 illustrates a side plan view of an exemplary embodiment of a surgical staple that may be implanted into bone in accordance with the present disclosure;
- (11) FIG. 7 illustrates a perspective view of an exemplary embodiment of a surgical staple comprised of a crown that includes an upper reinforced portion and a valley underneath the reinforced portion;
- (12) FIG. 8 illustrates a side plan view of an exemplary embodiment of a surgical staple that is indicated for fixation of osteotomies and joint arthrodeses of the hands and feet; and
- (13) FIG. 9 illustrates an upper perspective view of an exemplary embodiment of a surgical staple that may be implanted into bone in accordance with the present disclosure.
- (14) While the present disclosure is subject to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and will herein be described in detail. The invention should be understood to not be limited to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure.

DETAILED DESCRIPTION

(15) In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. It will be apparent, however, to one of ordinary skill in the art that the invention disclosed herein may be practiced without these specific details. In other instances, specific numeric references such as “first implant,” may be made. However, the specific numeric reference should not be interpreted as a literal sequential order but rather interpreted that the “first implant” is different than a “second implant.” Thus, the specific details set forth are merely exemplary. The specific details may be varied from and still be contemplated to be within the spirit and scope of the present disclosure. The term “coupled” is defined as meaning connected either directly to the component or indirectly to the component through another component. Further, as used herein, the terms “about,” “approximately,” or “substantially” for any numerical values or ranges indicate a suitable dimensional tolerance that allows the part or collection of components to function for its intended purpose as described herein.

(16) In general, the present disclosure describes an apparatus and a method for a surgical bending instrument for bending surgical implants. The surgical bending instrument comprises a body including a longitudinally extending hole, at least of portion of which comprising threads. A shaft comprising a threaded portion passes through hole and is rotatably engaged with the threads in the hole. A handle is fixedly coupled to a proximal end of the shaft, and a distal extension of the shaft

comprises a driver. A distal force applicator comprises a first grip, a second grip, and the driver centered between the first and second grips. In some embodiments, the distal force applicator is configured to retain a surgical staple, such that the surgical staple may be changed to a distracted configuration suitable for implantation at a bone fixation or fusion site of a patient. In some embodiments, the distal force applicator is configured to bend a bone fusion plate so as to tailor the plate to specific anatomy of the patient's bone.

(17) FIG. 1 illustrates a perspective view of an exemplary embodiment of a surgical bending instrument **100** in accordance with the present disclosure. The surgical bending instrument **100** comprises a proximal handle **104**, a body **108**, and a distal force applicator **112**. In the embodiment illustrated in FIGS. 1-2, a surgical staple **116** is shown loaded within the distal force applicator **112**, such that the surgical staple **116** is in a distracted configuration suitable for implantation at a bone fixation or fusion site of a patient. The surgical staple **116** generally is of the variety indicated for fixation of osteotomies and joint arthrodeses of the hands and feet. As such, the surgical staple **116** preferably is comprised of a metal alloy exhibiting shape memory and superelastic properties, such as Nitinol or other similar material. It should be understood, however, that the surgical bending instrument **100** is not to be limited to distracting surgical staples, but rather may be used in various other capacities, such as by way of non-limiting example, bending a bone fusion plate so as to tailor the plate to specific anatomy of a patient's bone being treated.

(18) As best illustrated in FIG. 2, the distal force applicator **112** comprises a first grip **120**, a second grip **124**, and a driver **128** centered there between. The first and second grips **120**, **124** are configured to cooperate so as to support a crown **132** of the surgical staple **116** when the driver **128** is placed into forcible contact with the center of the crown **132**. The forces on the crown **132** due to the driver **128** and the grips **120**, **124** operate to retain the surgical staple **116** within the distal force applicator **112**, thereby facilitating implanting the staple into the patient. As shown in FIG. 2, the first and second grips **120**, **124** are configured to engage the surgical staple **116** on opposite sides of the crown **132**. As will be appreciated, engaging the crown **132** on opposite sides advantageously prevents the staple from becoming dislodged from the distal forced applicator **112** during distracting the staple or implantation into the patient.

(19) Preferably, the first and second grips **120**, **124** have a separation distance comparable with the length of the crown **132**. Thus, as illustrated in FIGS. 3A-3B, the first grip **120** is positioned near a first leg **136** of the surgical staple **132**, and the second grip **124** is positioned near a second leg **140** of the staple. It will be appreciated, therefore, that the distal force applicator **112** of the surgical bending instrument **100** is configured to be used with specific sizes of surgical staples **116**. In some embodiments, the surgical bending instrument **100** may be configured with an overall size so as to be used with specific sizes of surgical staples **116**. In some embodiments, a color-matching system may be utilized to indicate to the surgeon that certain surgical staples **116** may be implanted by way of the surgical bending instrument **100**. Further, in some embodiments, the first and second grips **120**, **124** may have an adjustable separation distance so as to facilitate using the surgical bending instrument **100** with a variety of differently-sized staples.

(20) FIGS. 3A and 3B illustrate ghost views of the exemplary embodiment of the surgical bending instrument **100** illustrated in FIGS. 1-2. A shaft **144** passing through the body **108** comprises a threaded portion **148** and the above-mentioned driver **128**. In the illustrated embodiment, the driver **128** comprises a distal extension of the shaft **144**, having a narrower diameter than proximal portions of the shaft **144**. In some embodiments, however, the driver **128** and the shaft **144** are separate components which are engaged with one another. It will be appreciated that the shaft **144** communicates mechanical forces applied at the proximal handle **104** to the driver **128**, and thus to the crown **132** of the staple. In the illustrated embodiment, the threaded portion **148** is rotatably engaged with similar threads within a hole **152** extending longitudinally through the body **108**. The shaft **144** is fixedly coupled with the proximal handle **104** such that rotating the proximal handle **104** relative to the body **108** rotates the shaft **144** within the hole **152**, and thus moves the driver

128 longitudinally relative to the body **108**.

(21) As best illustrated in FIG. 3A, when the driver **128** is not in contact with the crown, the surgical staple **116** is in an initial configuration wherein the first leg **136** and the second leg **140** of the staple bend toward one another. During operation of the surgical bending instrument **100**, twisting the proximal handle **104** clockwise, as viewed from the perspective of a surgeon operating the instrument **100**, advances the shaft **144** and moves the driver **128** distally toward the crown **132**. Once the driver **128** contacts the center of the crown **132**, the surgical staple **116** becomes clasped between the grips **120**, **124** and the driver **128**. Further clockwise twisting of the proximal handle **104** exerts an increasing degree of force onto the crown **132** by the grips **120**, **124** and the driver **128**. As will be appreciated, the distally-directed force due to the driver **128** and proximally-directed forces due to the grips **120**, **124** cause a corresponding distraction of the first and second legs **136**, **140**. Thus, clockwise twisting of the proximal handle **104** enables the surgeon to change the surgical staple **116** from the initial configuration, shown in FIG. 3A, to a distracted configuration wherein the first and second legs **136**, **140** are forced into a parallel relationship, as illustrated in FIG. 3B.

(22) Once the surgical staple **116** is in the distracted configuration, the surgeon may use the surgical bending instrument **100** to slide the first and second legs **136**, **140** into parallel holes drilled across a bone fusion or fixation site of a patient. It will be appreciated that the threaded portion **148** keeps the driver **128** in contact with the crown **132**, thereby maintaining the distracted configuration of the surgical staple **116** during implantation. The surgeon may insert the first and second legs **136**, **140** into the parallel holes until the first and second grips **120**, **124** come into contact with the patient's bone. The surgeon may then twist the proximal handle **104** counterclockwise so as to retract the driver **128** proximally away from the crown **132**, thereby allowing the surgical staple **116** to relax from the distracted configuration. The shape memory and superelastic properties of the material comprising the surgical staple **116** ensure that the staple advantageously compresses the bones to be fused as the staple attempts to return to the initial configuration illustrated in FIG. 3A. Once the driver **128** has been suitably retracted from the crown **132**, the surgeon may disengage the first and second grips **120**, **124** from the crown and then use a tamp to push the legs **136**, **140** into the parallel holes until the crown **132** is in direct contact with the bone. The surgeon may then close the incision by way of suturing.

(23) As will be recognized, some surgical procedures may necessitate removing a surgical staple, such as the surgical staple **116**, from a bone fusion or fixation site of a patient. It is envisioned that the surgeon may lift the crown **132** away from the patient's bone surface by way of a suitable removal tool so as to create enough clearance for the surgeon to engage the grips **120**, **124** between the bone and the crown **132**. With the first and second grips **120**, **124** suitably engaged with the crown **132**, the surgeon may turn the proximal handle **104** clockwise to distally advance the driver **128** into contact with the center of the crown **132**. Upon twisting the proximal handle **104** so as to place the surgical staple **116** into the distracted configuration illustrated in FIG. 3A, the surgeon may use the surgical bending instrument **100** to pull the first and second legs **136**, **140** free of the patient's bone.

(24) FIGS. 4A and 4B illustrate ghost views of an exemplary embodiment of a surgical bending instrument **160** retaining the surgical staple **116** in accordance with the present disclosure. The surgical bending instrument **160** illustrated in FIGS. 4A-4B is similar to the surgical bending instrument **100** illustrated in FIGS. 1-2, with the exception that the surgical bending instrument **160** comprises a side lever **164** coupled with a driver **168**. The side lever **164** comprises a wheel **172** which is rotatably mounted on a pivot **176** within a body **180** of the surgical bending instrument **160**. The wheel **172** is in sliding contact with a proximal portion of the driver **168**, such that the side lever **164** may be rotated from a distal position, illustrated in FIG. 4A, to a proximal position as shown in FIG. 4B. The wheel **172** further comprises a cam **184** configured to push the driver **168** distally within the body **180** into the crown **132** when the side lever **164** is rotated to the proximal

position. As will be appreciated, the cam **184** and the driver **168** exert a distally-directed force onto the crown **132**, thereby distracting the surgical staple **116**, as illustrated in FIG. **4B** and described herein with respect to FIGS. **3A** and **3B**.

(25) It will be appreciated that moving the side lever **164** to distract the surgical staple **116** requires a degree of force to be placed onto the lever. A proximal surface **188** of the body **180** advantageously facilitates grasping and stabilizing the surgical bending instrument **160** while the side lever **164** is moved during distraction of the surgical staple **116**. A narrow midsection **192** of the body **180** further enables grasping and stabilizing the surgical bending instrument **160** during distraction of the staple **116**.

(26) FIGS. **5A** and **5B** illustrate proximal views of an exemplary embodiment of a surgical bending instrument **196** retaining the surgical staple **116** in accordance with the present disclosure. The surgical bending instrument **196** illustrated in FIGS. **5A** and **5B** is similar to the surgical bending instrument **160** illustrated in FIGS. **4A-4B**, with the exception that the instrument **196** comprises a center lever **200** rotatably positioned within an opening **204** of a body **208**. As best illustrated in FIG. **5A**, the center lever **200** comprises a wheel **212** and a cam **216** positioned on a periphery of the wheel. The wheel **212** is rotatably mounted on a pivot **220** within the body **208**. Unlike previously described embodiments, in the embodiment of FIGS. **5A-5B**, the wheel **212** is in direct contact with the crown **132**, in absence of a separately, coupled driver. In the embodiment illustrated in FIGS. **5A-5B**, the cam **216** operates substantially similarly to the driver described herein. It will be appreciated that the cam **216** is positioned on the wheel so as to exert a distally-directed force on the crown **132** when the center lever **200** is moved from a distal position, illustrated in FIG. **5A**, to a proximal position shown in FIG. **5B**. As described herein, the distally-directed force on the crown **132** distracts the surgical staple **116** such that the staple may be implanted in the bone of the patient.

(27) As best illustrated in FIG. **5B**, the center lever **200** further comprises a slot **224** adjacent to the wheel **212**. It will be appreciated that the slot **224** advantageously allows the center lever **200** to be placed into the distal position while the surgical staple **116** is clasped in the first and second grips **120**, **124**. In some embodiments, the slot **224** is configured to cooperate with the first and second grips **120**, **124** so as to retain the surgical staple **116** in the surgical bending instrument while the staple is in the initial configuration, as shown in FIG. **5A**.

(28) It should be understood that although embodiments of the surgical bending instrument have been discussed in combination with the surgical staple **116**, the surgical bending instruments **100**, **160**, **196** are not to be limited to distracting surgical staples. Rather, the surgical bending instruments **100**, **160**, **196** may be used in various capacities other than as described herein, such as by way of non-limiting example, bending a bone fusion plate so as to tailor the plate to a specific anatomy of a patient's bone being treated. Accordingly, it is envisioned that the surgical bending instruments **100**, **160**, **196** may be packaged into sterile surgery-specific kits comprising other surgical tools and components, such as by way of non-limiting example, drill guides, drill sizers, tamps, forceps, staple removal tools, drills, temporary pins, drill depth stops, fusion bone plates, bone plate fasteners, compression screws, and the like. Further, the surgical bending instruments **100**, **160**, **196** preferably comprise a rigid material suitable for bending surgical implants, such as surgical staples and bone fusion plates, as described herein. In some embodiments, the surgical bending instruments **100**, **160**, **196** comprise metal, plastic, or a combination of the two.

(29) Embodiments of the surgical bending instruments **100**, **160**, **196** have been described in combination with the surgical staple **116**. It is to be understood, however, that the surgical staple **116** is not to be limited to the particular embodiment of the surgical staple described in connection with FIGS. **1-5B**. As such, it is contemplated that in some embodiments, multiple teeth suitable for engaging with bone may be disposed on one or more surfaces of the surgical staple **116**, such as, for example, along a backspan or crown of the staple. In the embodiment of the surgical staple **116**, for example, a multiplicity of teeth **240** are disposed along an inner side of each of the legs **136**,

140 of the surgical staple **116** so as to discourage movement of the staple once implanted in bone. The teeth **240** may be biased toward the crown **132** so as to facilitate inserting the staple into bone, while discouraging the staple from backing out of the bone after implantation.

(30) FIG. **6** illustrates a side plan view of an exemplary embodiment of a surgical staple **244** that may be implanted into bone in accordance with the present disclosure. The surgical staple **244** is similar to the surgical staple **116**, and thus the surgical staple **244** generally is of the variety indicated for fixation of osteotomies and joint arthrodeses of the hands and feet. In some embodiments, the surgical staple **244** may be comprised of a material that is amenable to being placed into the distracted configuration and relaxed therefrom, such as by way of any of the surgical bending instruments **100**, **160**, **196**. To this end, the surgical staple **244** may be comprised of a metal alloy that exhibits shape memory and superelastic properties, such as Nitinol or other similar material.

(31) As shown in FIG. **6**, multiple teeth **248** are disposed along an inner side of the first and second legs **136**, **140**. The teeth **248** are biased toward the crown **132**, such that the teeth **248** engage with bone and prevent the staple from backing out of the bone after implantation, yet the teeth **248** offer relatively little resistance during implantation of the surgical staple **244** into the bone. Although seven of the teeth **248** are shown disposed along each of the legs **136**, **140**, it is contemplated that more or less than seven teeth **248** may be incorporated along the legs **136**, **140** in various other embodiments of the surgical staple **244**. In addition to the teeth **248**, multiple exterior teeth **252** may be disposed along the first and second legs **136**, **140**. As shown in FIG. **6**, the surgical staple **244** includes two exterior teeth **252** disposed on an outside of each of the legs **136**, **140**, near the crown **132**. The exterior teeth **252** cooperate with the teeth **248** to engage the bone and prevent movement of the surgical staple **244** after being implanted into the bone.

(32) Moreover, teeth similar to the teeth **248**, **252** are not limited to being disposed along the first and second legs **136**, **140**, but rather teeth, as well as various other advantageous structures, may be incorporated into the crown **132**, without limitation. For example, in the embodiment illustrated in FIG. **6**, two teeth **256** are disposed on the lower surface of the crown **132**. The teeth **256** are biased toward a centerline of the crown **132** so as to engage with bone and prevent lateral movement of the crown **132** along the surface of the bone. In another example, FIG. **7** illustrates an embodiment of a surgical staple **260** comprised of a crown **264** that includes an upper reinforced portion **268** and a valley **272** underneath the reinforced portion. In the illustrated embodiment, the valley **272** is centrally disposed along the lower surface of the crown **264** and bordered by edges **276**. As will be appreciated, the edges **276** are configured to engage with bone and prevent lateral movement of the crown **264** along the surface of the bone once the surgical staple **260** is implanted. The reinforced portion **268** imparts structural integrity to the surgical staple **260** in compensation for the presence of the valley **264**.

(33) FIG. **8** illustrates a side plan view of an exemplary embodiment of a surgical staple **280** that is indicated for fixation of osteotomies and joint arthrodeses of the hands and feet. The surgical staple **280** is similar to the surgical staple **244**, illustrated in FIG. **6**, and thus the surgical staple **280** is comprised of a first leg **136** and a second leg **240** that both include inner teeth **248** and exterior teeth **252**, as described hereinabove. Unlike the surgical staple **244**, however, the surgical staple **280** is comprised of a crown **284** that includes a wing portion **288** that extends beyond each of the legs **136**, **140**. Disposed at opposite ends of the crown **284**, the wing portions **288** cooperate with the teeth **248**, **252**, as well as with the center portion of the crown **284**, to fixate the surgical staple **280** once implanted in bone.

(34) It should be understood that the wing portions **288** are not to be limited to being incorporated into the illustrated embodiment of the surgical staple **280**, shown in FIG. **8**, but rather the wing portions **288** may be practiced in combination with any of various features disclosed in connection with the surgical staples **244**, **260**. For example, in some embodiments, the wing portions **288** may be incorporated into the crown **132** comprising the surgical staple **244**. Thus, the wing portions **288**

and the teeth **252** may cooperate to fixate the surgical staple **244** once implanted into bone. Further, in some embodiments, the wing portions **288** may be combined with the upper reinforced portion **268** and the valley **272** to fixate the surgical staple **260** implanted in bone. It is further contemplated, therefore, that in some embodiments, the wing portions **288** may be practiced in absence of any or all of the teeth **248**, **252**, the reinforced portion **268**, and the valley **272**, without limitation.

(35) In some embodiments, the wing portions **288** may be configured to receive the first and second grips **120**, **124** of the surgical bending instrument **100** so as to place the surgical staple **280** into the distracted configuration, as described herein. For example, the first and second grips **120**, **124** may be configured to cooperate so as to support the wing portions **288** of the crown **284** when the driver **128** is placed into forcible contact with the center of the crown **284**. The force on the crown **284** due to the driver **128** and the force on the wing portions **288** due to the grips **120**, **124** operate to retain the surgical staple **280** within the distal force applicator **112**, thereby facilitating implanting the staple into the patient. As will be appreciated, engaging the wing portions **288** on opposite sides of the crown **284** advantageously prevents the surgical staple **280** from becoming dislodged from the distal forced applicator **112** during distracting the staple or implantation into the patient.

(36) FIG. **9** illustrates an upper perspective view of an exemplary embodiment of a surgical staple **292** that may be implanted into bone in accordance with the present disclosure. The surgical staple **292** generally is indicated for fixation of osteotomies and joint arthrodeses of the hands and feet. In some embodiments, the surgical staple **292** may be comprised of a material that is amenable to being placed into the distracted configuration and relaxed therefrom, such as by way of any of the surgical bending instruments **100**, **160**, **196**. Similar to the surgical staples discussed with respect to FIGS. **2-8**, the surgical staple **292** may be comprised of a metal alloy that exhibits shape memory and superelastic properties, such as Nitinol or other similar material.

(37) As will be appreciated, the surgical staple **292** is similar to the surgical staple **244**, illustrated in FIG. **6**, with the exception that the surgical staple **292** is comprised of four legs in lieu of the first and second legs **136**, **140** comprising the surgical staple **244**. As shown in FIG. **9**, the surgical staple **292** is comprised of a first transverse portion **296** disposed at one end of a crown **300** and a second transverse portion **304** disposed at an opposite end of the crown **300**. A first leg **308** and a second leg **312** extend downward and parallelly from opposite ends of the first transverse portion **296**. Similarly, a third leg **316** and a fourth leg **320** extend downward and parallelly from opposing ends of the second transverse portion **304**. The first and second legs **308**, **312** angle respectively toward the third and fourth legs **316**, **320**, giving the surgical staple **292** a resemblance of two of the surgical staples **244** coupled together in parallel. Thus, the first leg **308** and the third leg **316** cooperate similarly to the legs **136**, **140** of a first surgical staple **244**, and the second leg **312** and the fourth leg **320** cooperate similarly to the legs **136**, **140** of a second surgical staple **244**. As such, the surgical staple **292** is particularly well suited for implantation into four holes that are drilled in bone across a bone fusion site.

(38) It is contemplated that any of the legs **308**, **312**, **316**, **320** may be practiced in combination with any of the various features disclosed hereinabove in connection with the surgical staples **244**, **260**, **280**, without limitation. For example, in the illustrated embodiment of FIG. **9**, each of the legs **308**, **312**, **316**, **320** includes multiple inner teeth **248** and a pair of exterior teeth **252**, as described herein. The teeth **248** disposed on the first and second legs **308**, **312** are oriented toward the teeth **248** disposed on the third and fourth legs **316**, **320**. Thus, once the surgical staple **292** is implanted into bone and allowed to relax from the distracted configuration, the legs **308**, **312**, **316**, **320** cooperatively compress the bones to be fused as the staple attempts to return to the initial configuration illustrated in FIG. **9**. The teeth **248** engage with the bone and ensure that the implanted surgical staple **292** remains fixated. The exterior teeth **252** operate similarly to the teeth **248**, engaging with the bone and fixating the surgical staple **292**, as disclosed herein.

(39) It is contemplated that the surgical staple 292 may be practiced in combination with any of various features disclosed in connection with the surgical staples 244, 260, 280, without limitation. For example, in some embodiments, the wing portions 288 may be incorporated into the crown 300. In some embodiments, wing portions 288 may be coupled with the opposite ends of each of the first and second transverse portions 296, 304. It is envisioned, therefore, that the wing portions 288 may cooperate with the teeth 248, 252 to fixate the surgical staple 292 once implanted into bone. Further, in some embodiments, the wing portions 288 may be incorporated into the crown 300 in combination with the upper reinforced portion 268 and the valley 272 to fixate the surgical staple 260 implanted in bone. Moreover, in some embodiments, the wing portions 288 may be incorporated into the surgical staple 292 in lieu of any or all of the teeth 248, 252, the reinforced portion 268, and the valley 272, without limitation.

(40) While the invention has been described in terms of particular variations and illustrative figures, those of ordinary skill in the art will recognize that the invention is not limited to the variations or figures described. In addition, where methods and steps described above indicate certain events occurring in certain order, those of ordinary skill in the art will recognize that the ordering of certain steps may be modified and that such modifications are in accordance with the variations of the invention. Additionally, certain of the steps may be performed concurrently in a parallel process when possible, as well as performed sequentially as described above. To the extent there are variations of the invention, which are within the spirit of the disclosure or equivalent to the inventions found in the claims, it is the intent that this patent will cover those variations as well. Therefore, the present disclosure is to be understood as not limited by the specific embodiments described herein, but only by scope of the appended claims.

Claims

1. A surgical bending instrument for implanting a surgical staple at a bone fusion site of a patient, comprising: a body having a distal force applicator and a proximal handle; a first grip and a second grip comprising the distal force applicator wherein the first grip and the second grip comprise oppositely oriented L shaped forms; and a driver in mechanical communication with the proximal handle.
2. The surgical bending instrument of claim 1, wherein the first grip and the second grip are separated by a distance comparable with a length of a crown comprising the surgical staple.
3. The surgical bending instrument of claim 2, wherein the first grip is configured to be positioned near a first leg of the surgical staple while the second grip is configured to be positioned near a second leg of the surgical staple.
4. The surgical bending instrument of claim 2, wherein the distal force applicator is configured to provide an adjustable separation distance between the first grip and the second grip.
5. The surgical bending instrument of claim 1, wherein the distal force applicator is configured to be used with specific sizes of surgical staples.
6. The surgical bending instrument of claim 1, wherein any one or more of the body, the distal force applicator, and the proximal handle are configured with an overall size so as to be used with specific sizes of the surgical staple.
7. The surgical bending instrument of claim 6, wherein any one or more of the body, the distal force applicator, and the proximal handle include a color-matching system so as to indicate those certain surgical staples that are implantable by way of the surgical bending instrument.
8. The surgical bending instrument of claim 1, wherein a shaft is disposed inside the body and extends from the proximal handle to the distal force applicator.
9. The surgical bending instrument of claim 8, wherein the driver comprises a distal extension of the shaft that is configured to forcibly contact a crown comprising the surgical staple.
10. The surgical bending instrument of claim 9, wherein the shaft includes a threaded portion

configured to advance the driver into the crown when the proximal handle is turned with respect to the body.
